Inference using normal distributions

Overview

Very quick review of hypothesis tests for more than 2 means and for correlation

Statistical inference using parametric probability distributions

The normal distribution

- Generating random data
- Visualizing the normal distribution
- Calculating probabilities from a normal distribution

Converting to the standard normal distribution

Announcement

Homework 8 has been posted!

It is due on Gradescope on Sunday April 5th at 11pm

We will discuss the final project next week

Very quick review

Very quick review: Hypothesis test for multiple means

Step 1:

$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for all i, j in {as, ns, ss, ah}

Step 2:

boxplot(quant_data ~ cat_data)

$$MAD = (|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

3. Create the null distribution!

Group1 | Group 2 | . . . | Group k

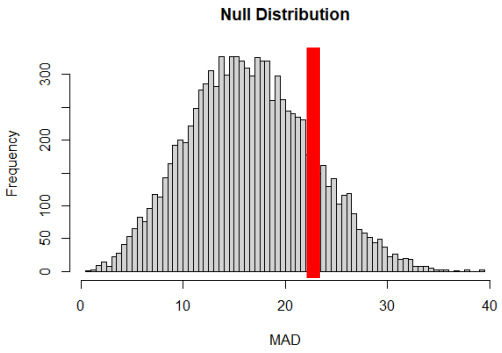
Reconstructed participant pool data under H_0

Shuffle data for random assignment consistent with H_0

Shuffled 'Group 1' | Shuffled 'Group 2' | . . . | Shuffled 'Group k'

Compute MAD statistic from shuffled groups

4. p-value



Very quick review: Hypothesis test for correlation

1. State the null and alternative hypothesis:

   H_0: $\rho = 0$

Possible alternative hypotheses:

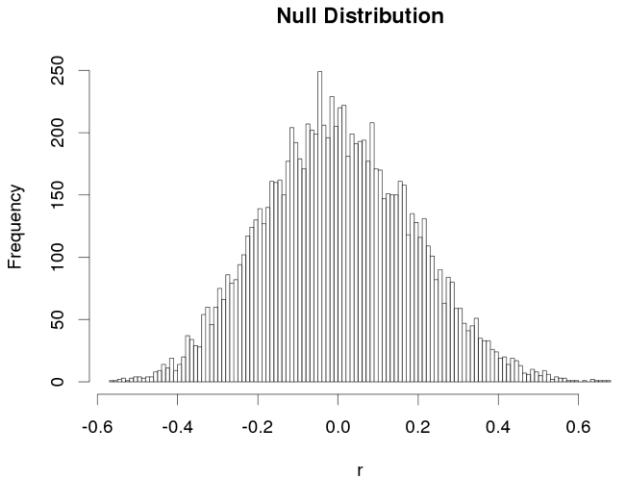
   a. H_A: $\rho < 0$
   b. H_A: $\rho > 0$
   c. H_A: $\rho \neq 0$

2. plot(x, y) and cor(x, y)

3. cor(x, shuffle(y))

4. p-value

5.



Statistical inference

Randomization/computational methods

vs.

Parametric probability functions

Big picture: statistical inference

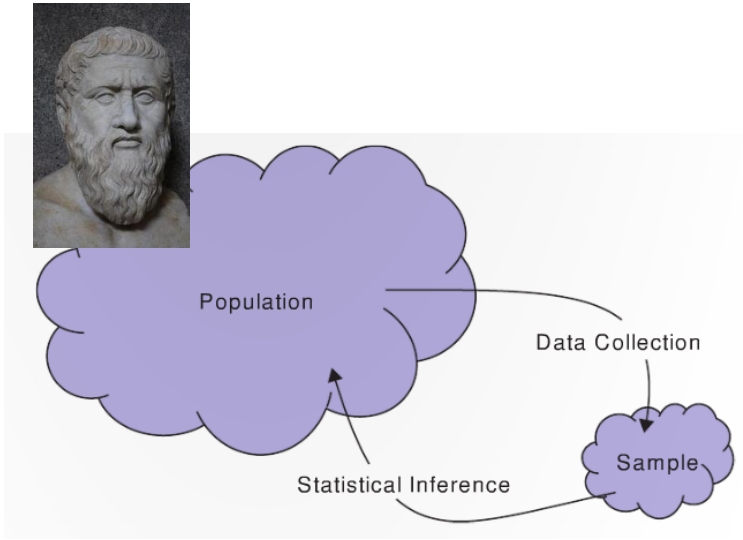
0.4 0.6

ρ

Statistic inference always involves using samples of data to make claims about a population

- i.e., using statistics to make claims about parameters

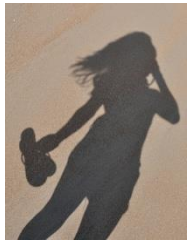
π, μ, σ, ρ



To quantify the **size of an effect**, we have created <u>confidence intervals</u>

- These intervals contain a parameter of interest most of the time

$\hat{p}, \bar{x}, s, r$



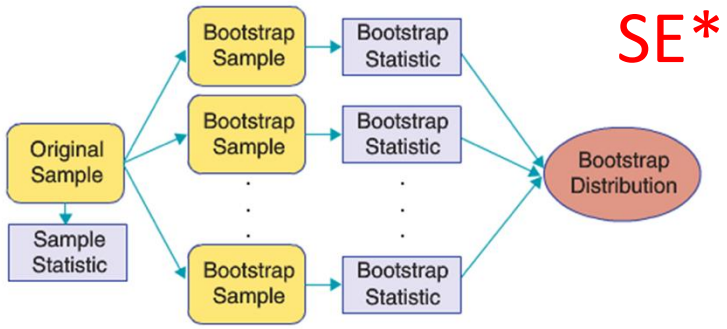
To assess if a particular **parameter value is plausible**, we have run <u>hypothesis tests</u>

$H_0: \rho = 0$

Inference using randomization/computational methods

So far, we have used computational (randomization) methods for inference

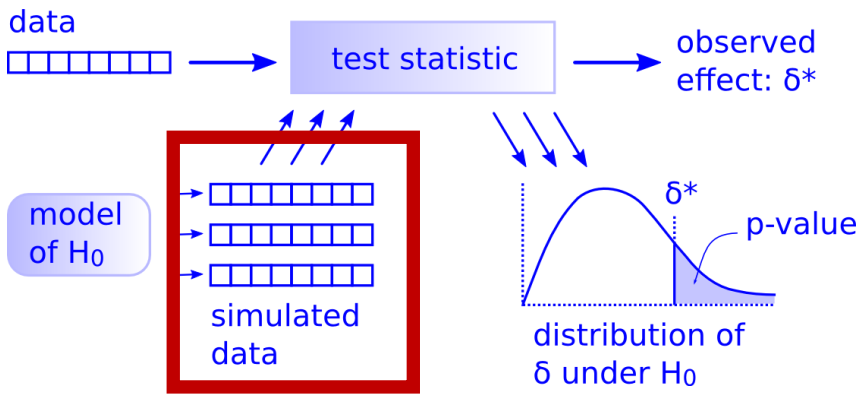
- i.e., we computationally repeat a process many times using the do_it() function



For confidence intervals, we used the bootstrap to repeatedly sample with replacement from our original sample

$$\text{CI} = \text{statistic} \pm q^* \cdot SE^*$$

For hypothesis tests, we have created null distributions by repeatedly generating statistics consistent with the null hypothesis

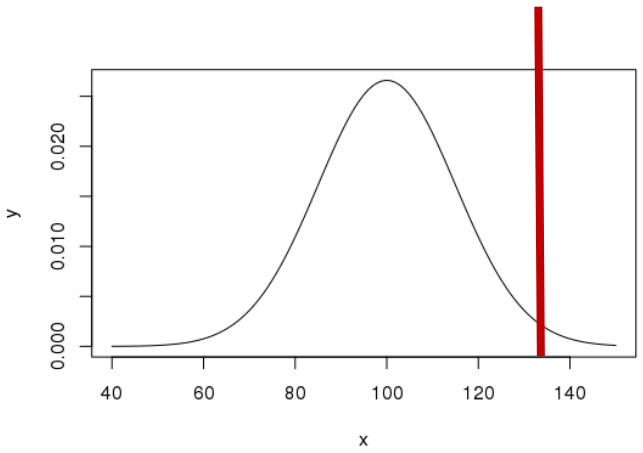


Inference using randomization/computational methods

Data	1 Sample	2 Samples	> 2 Samples
Categorical data	H_0: $\pi = p_0$ H_A: $\pi \neq p_0$ <u>Flip "coins"</u> rflip()	H_0: $\pi_1 = \pi_2$ H_A: $\pi_1 \neq \pi_2$ <u>Flip "coins"</u> rflip()	H_0: $\pi_1 = p_1$, $\pi_2 = p_2$, … , $\pi_k = p_k$ H_A: At least one p_i is different than specified <u>Roll a k-sided die n times</u> rroll(1, n, prob =)
Quantitative data	H_0: $\mu = v_0$ H_A: $\mu \neq v_0$ <u>resample</u> sample(… , replace = TRUE)	H_0: $\mu_1 = \mu_2$ H_A: $\mu_1 \neq \mu_2$ <u>Shuffle data</u> shuffle()	H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$ H_A: At least one μ_i is different <u>Shuffle data</u> shuffle()

Inference parametric probability models

We are going to do the same inference procedures but use mathematical functions instead of computational methods

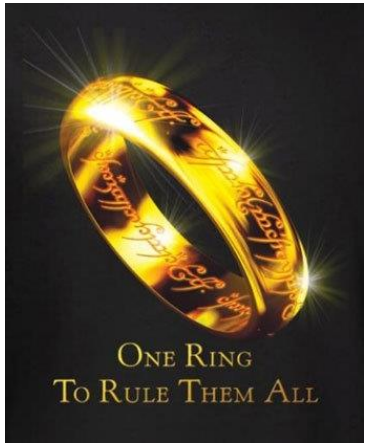


For confidence intervals, we will have formulas for SE estimates

- CI = stat ± q* $\cdot \hat{SE}$ $\qquad \hat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

For hypothesis tests, we will have mathematical functions for the null distribution

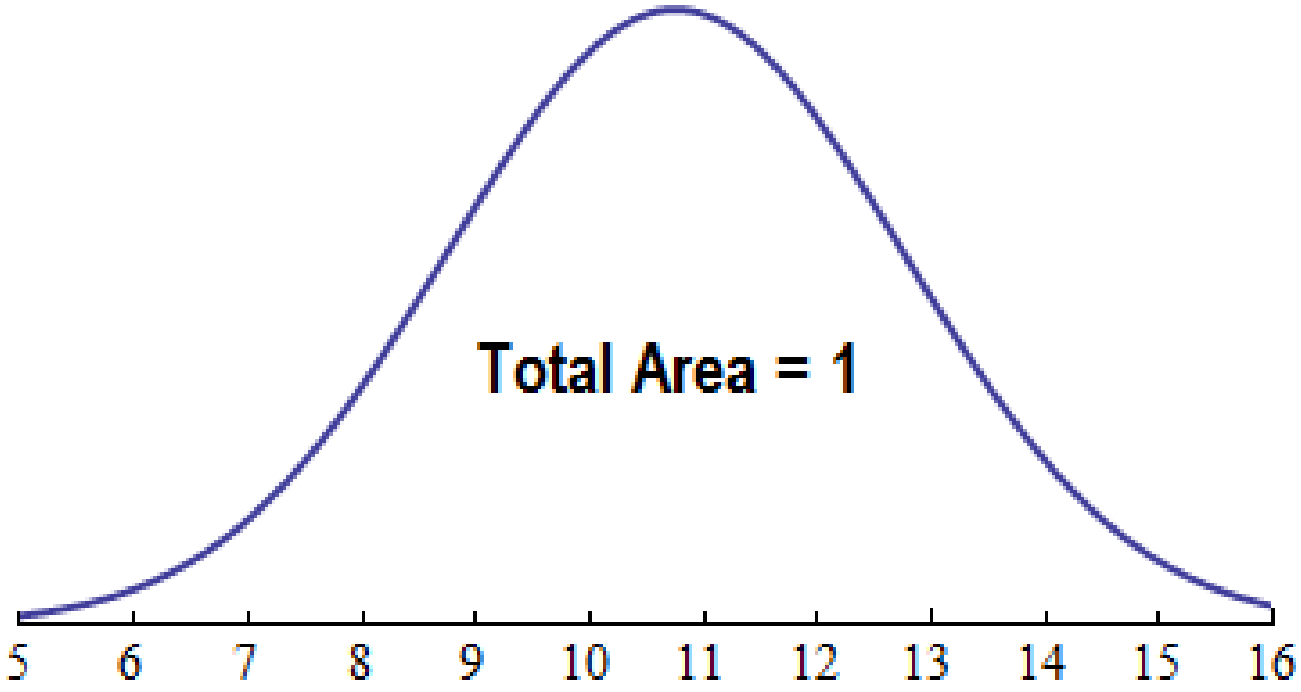
- E.g., our null distribution will be a normal distribution



But conceptually, everything is the same!

Let's get started!

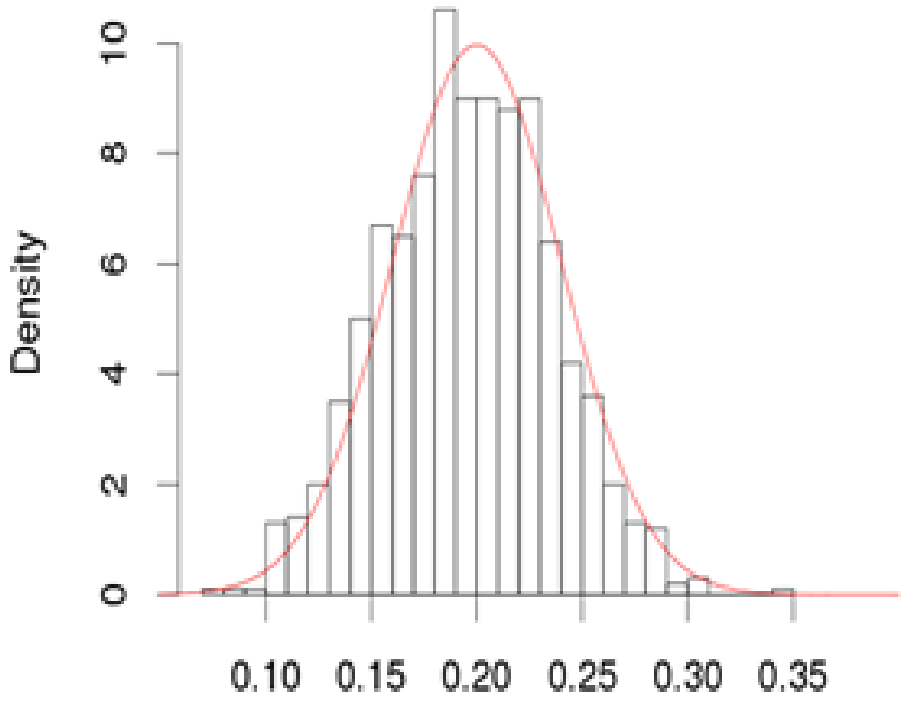
Inference using parametric probability distributions



Inference using parametric probability distributions

We can use mathematical functions called **probability distributions** to do statistical inference

For example, instead of running computer simulations to create null distributions, we can just mathematical probability distributions



Inference using parametric probability distributions

There are several functions that allow use to interact with probability distributions:

1. **Random number generation** functions (**r**)
2. **Density functions** to visual probability distributions (**d**)
3. **Cumulative probability distribution** functions to get the probability of events (**p**)

Note: Many probability distributions have small number of ***parameters*** that control the shape of the probability distribution

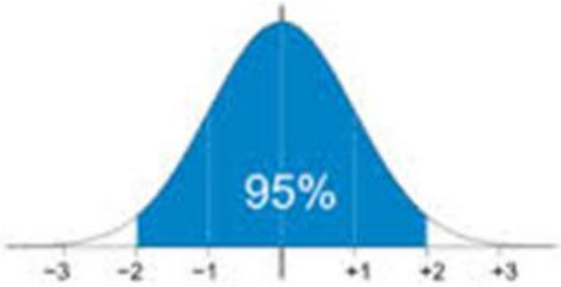
- Using these probability models for statistical inference is called “parametric statistics”

Main use case: Normal probability distributions

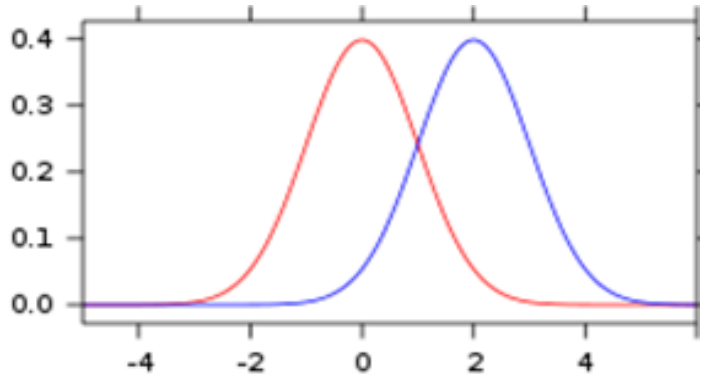
Normal distributions are a family of bell-shaped curves with two parameters

- The mean: μ
- The standard deviation: σ

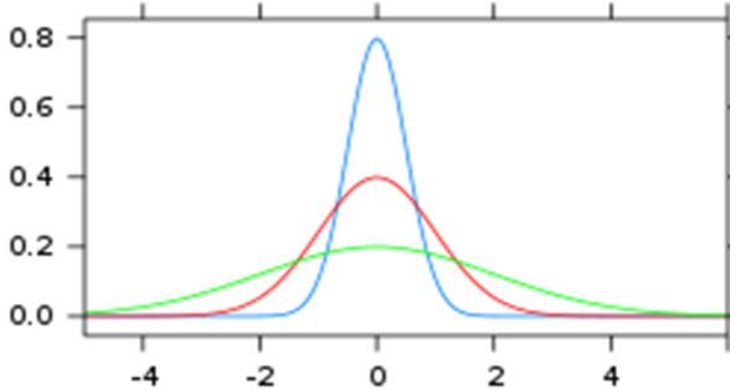
Once μ and σ are set, we have a specific probability distribution that we use to get probabilities of events, generate random data, etc.



Changing μ



Changing σ



1. Generating random numbers (rnorm)

Random numbers (i.e., random variables) that we have not yet observed are denoted with capital letters

- E.g., X, Z, etc.

We denote random numbers that come from a **normal distribution** using $X \sim N(\mu, \sigma)$

We can generate ***n*** random from a normal distribution using the rnorm(n, mean, sd)

Example: IQ scores

Intelligence quotient (IQ) is a score derived from a set of standardized tests to assess human intelligence

IQ scores are normally distributed and have a mean of 100 and a standard deviation of 15

Let's use the rnorm() to simulate some random IQ scores

- i.e., to generate values: $X \sim N(100, 15)$

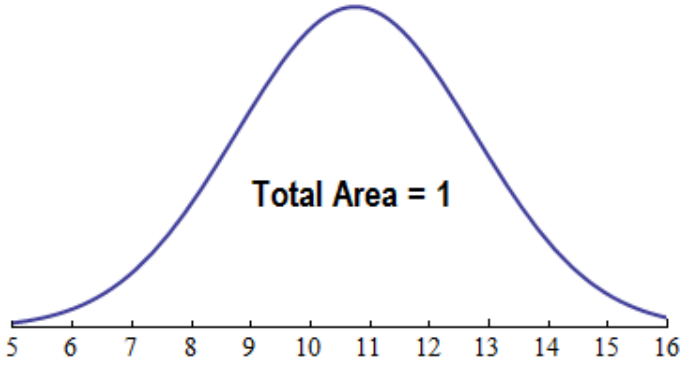
Let's try it in R!

2. Density Curves (dnorm)

A **density curve** is a mathematical function f(x) that can be used to calculate probabilities

Density curves have two defining properties:

1. The total area under the curve f(x) is equal to 1
2. The curve is always ≥ 0



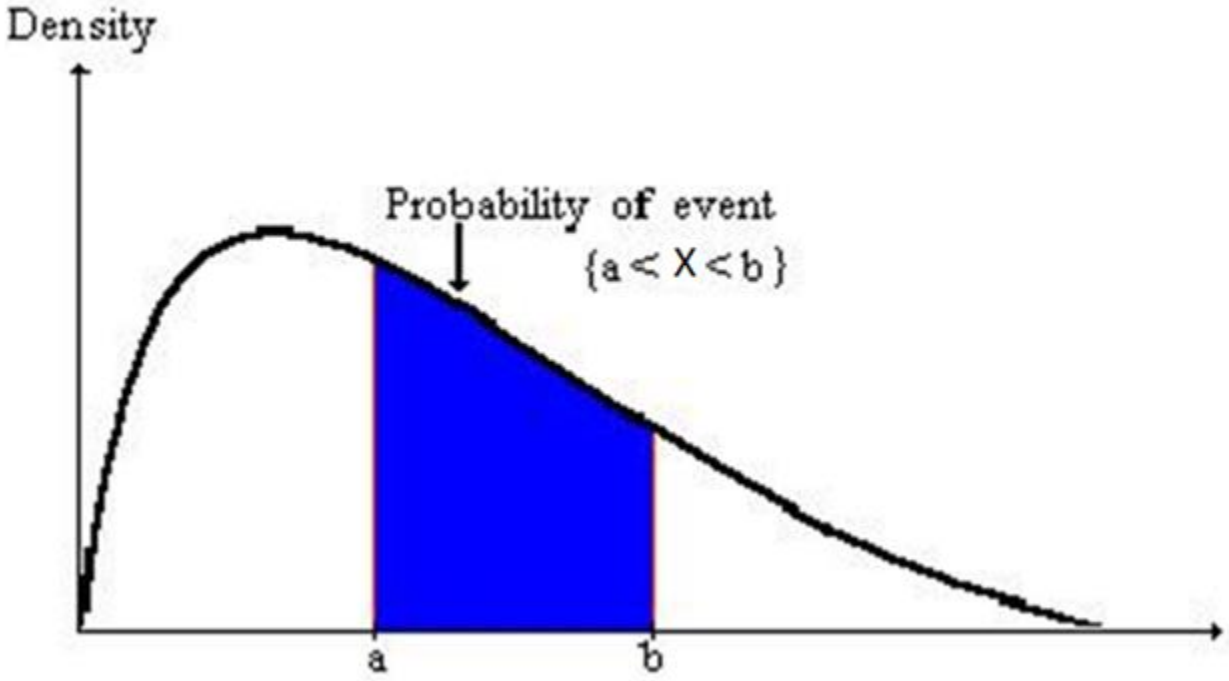
We can think of density curves as histograms that have:

- Infinitely large data sample
- With infinitely small bins sizes
- Normalized to have an area of 1

2. Density Curves

The **area under the density curve** in an interval [a, b] models the probability that a random number X will be in the interval

P(a < X < b) is the area under the curve from a to b



$$P(a < X < b) = \int_a^b f(x)dx$$

The Normal density function

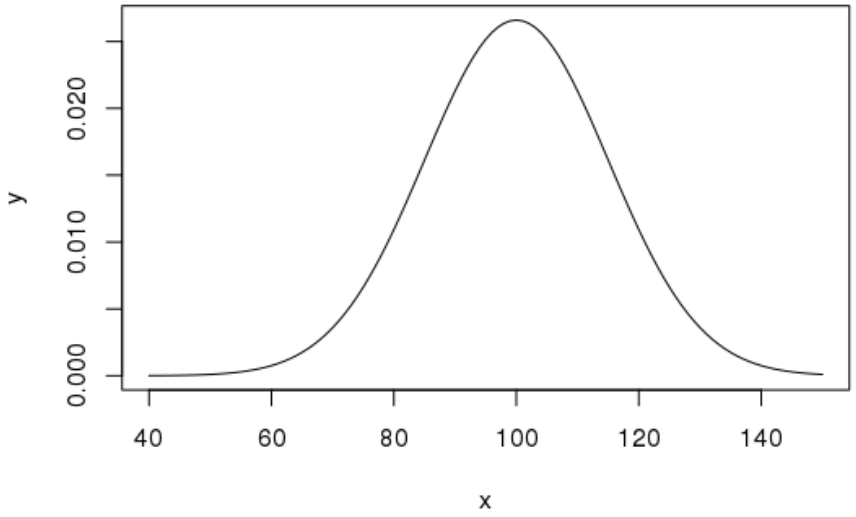
For normal distributions, the equation for the density function is:

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

R: dnorm(x, mean, sd)

In this class we will not be concerned with the details of this function

We just need to know that once μ and σ are set, it is just a regular function that takes in an x value and returns a y value



Graphing Normal Curves

We will use density functions to visualize which values are most likely

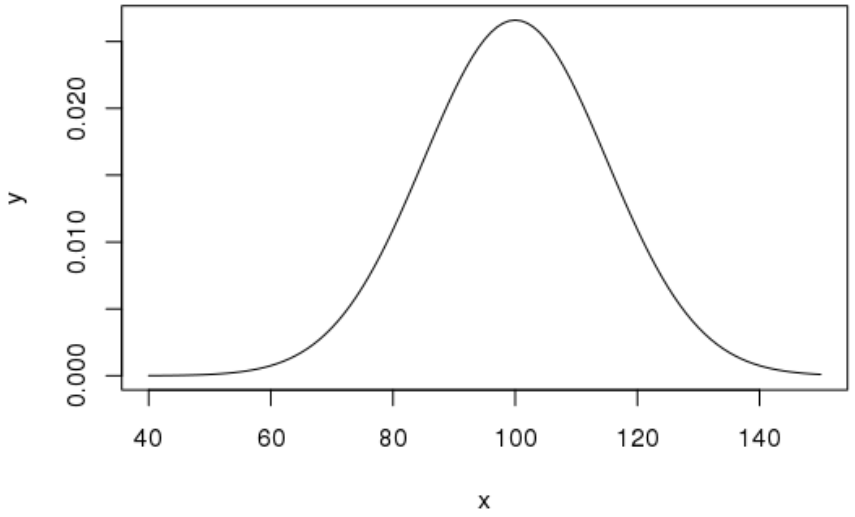
In R we will use the dnorm() function to do this

μ σ

```
x <- 40:150

y <- dnorm(x, 100, 15)

plot(x, y, type = "l")
```



Let's try it in R!

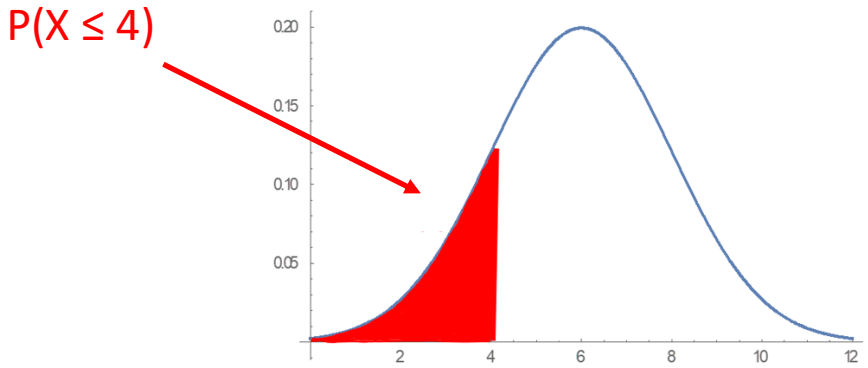
3. Cumulative distribution functions (pnorm)

Cumulative distribution functions give the probability of getting a random value X less than or equal to an input value x: $P(X \leq x)$

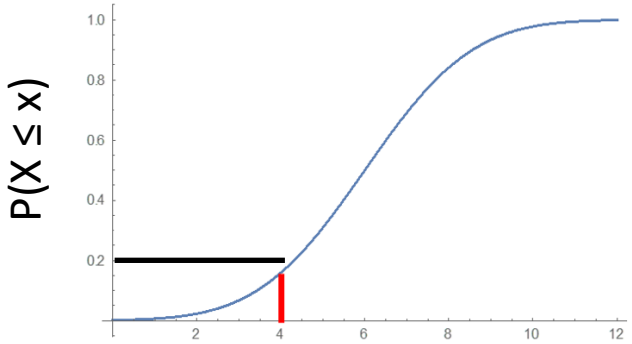
- For example, we would write the probability of getting a random number X less than 2 as: $P(X \leq 2)$

Cumulative distribution functions are obtained by calculating the area under a probability density function

Density: f(x)



Cumulative distribution F(x)



$$P(X \leq x)$$

$$= \text{pnorm(x)}$$

$$= \int_{-\infty}^{x} f(x)dx$$

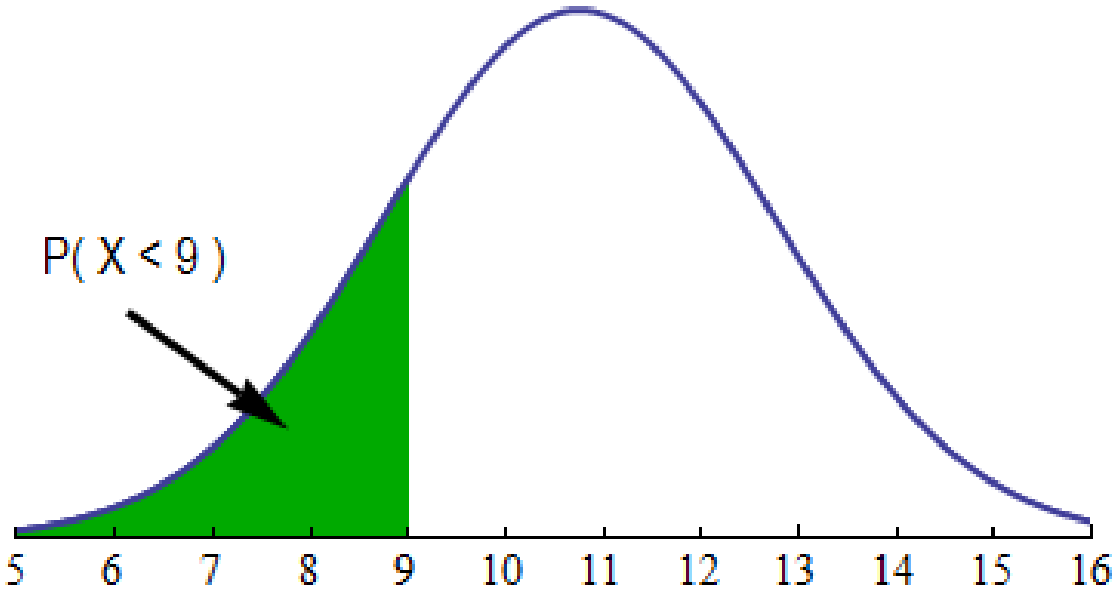
Finding normal probabilities of a normal curve

To get the probability (area) from a normal distribution we can use the pnorm() function

```
pnorm(x, mean, sd)
```

P(X < 9; 11, 3) μ σ

```
pnorm(9, 11, 3)
```



Let's try it in R!

Summary of R normal probability functions

Generate random data

- rnorm(m, mu, sigma)

Plot the density curve

- dnorm(x_vec, mu, sigma)

Get the probability that we would get a random value less than x

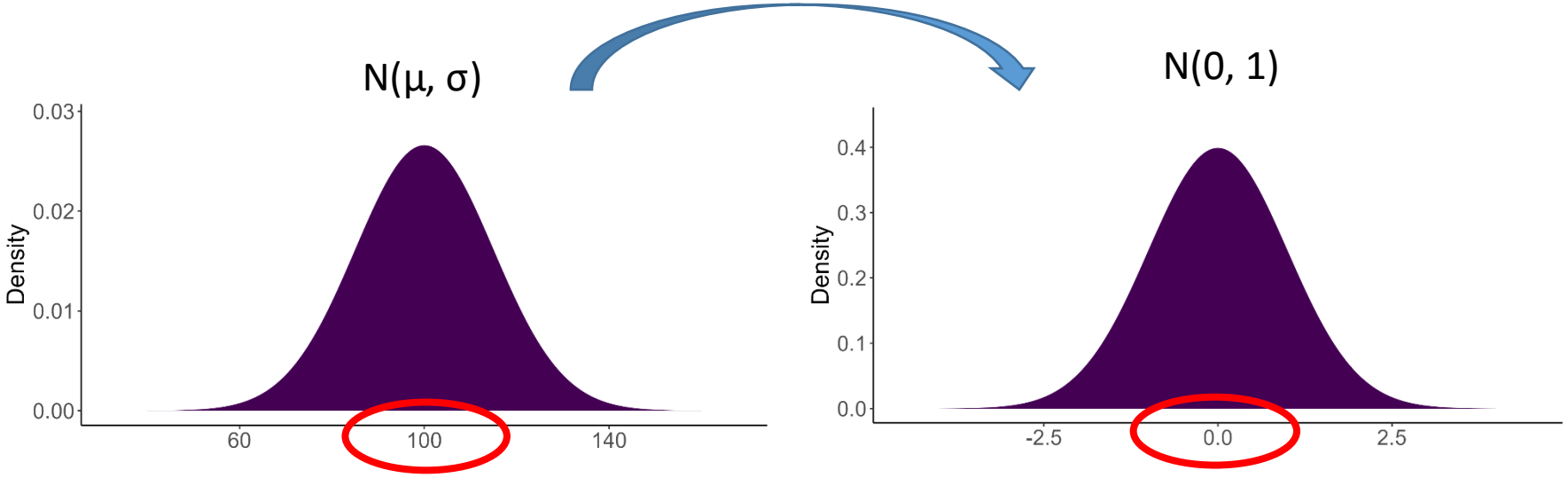
- pnorm(x_vec, mu, sigma)

The Standard Normal distribution

Standard Normal N(0, 1)

Since all normal distributions have the same shape, it is convenient to convert them to a standard scale with: **$\mu = 0$, $\sigma = 1$**

This is called the **standard normal** distribution: $Z \sim N(0, 1)$



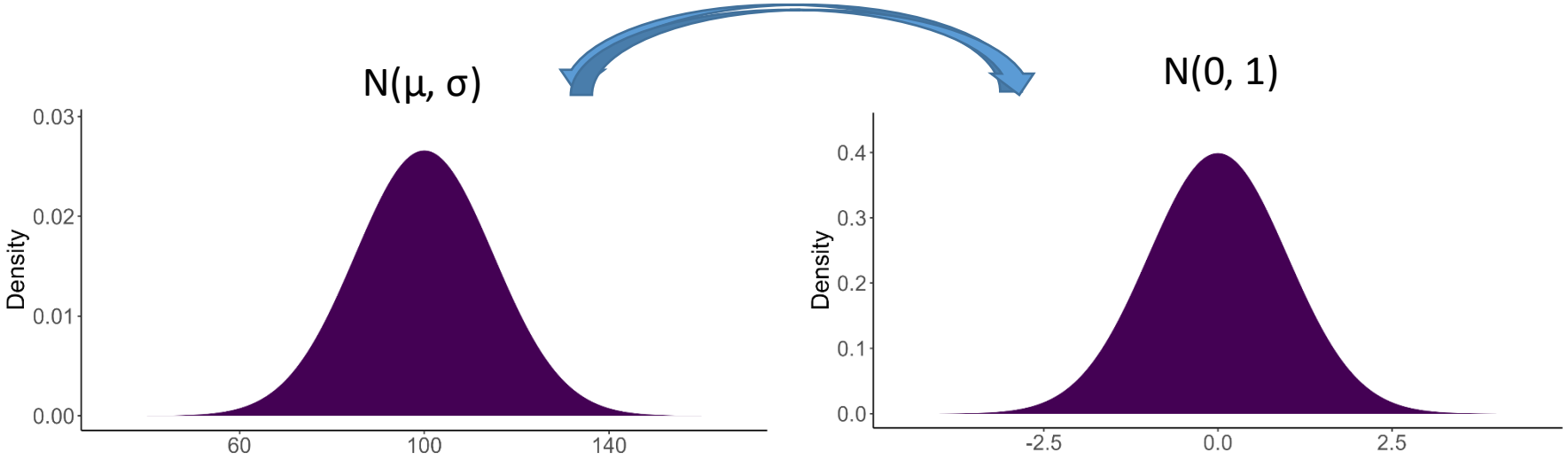
Converting to the standard normal distribution

We can apply a z-score transformation to any normally distributed random variable $X \sim N(\mu, \sigma)$ to convert it to the standard normal distribution $Z \sim N(0, 1)$:

$$Z = (X - \mu) / \sigma$$

To convert from $Z \sim N(0, 1)$ to any $X \sim N(\mu, \sigma)$, we reverse the standardization with:

$$X = \mu + Z \cdot \sigma$$



Converting to the standard normal distribution

1. What is the Z-score of someone who has an IQ score of 112?

$$Z = (X - \mu) / \sigma$$

2. What if someone has a Z-score of 2.2, what is their IQ score?

$$X = \mu + Z \cdot \sigma$$

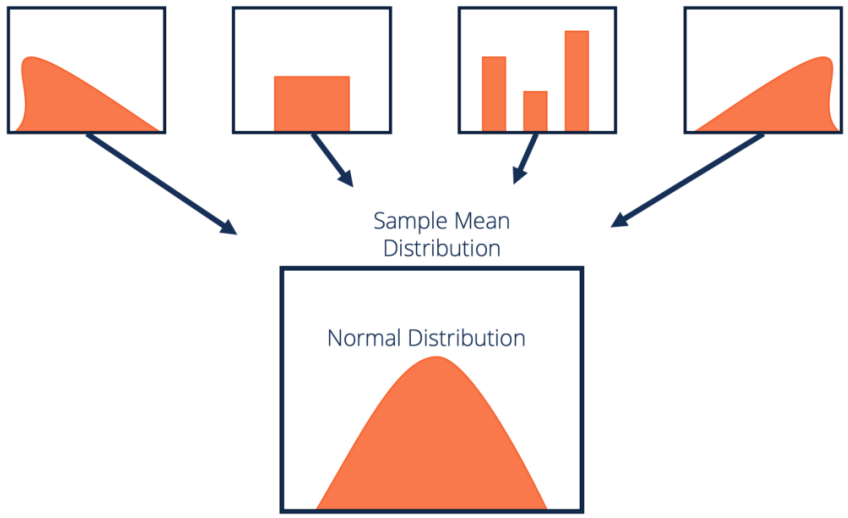
Let's quickly try it in R!

The Central Limit Theorem

Central limit theorem

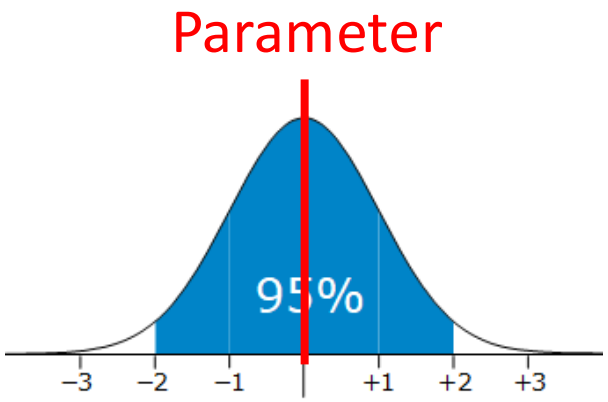
For random samples with a sufficiently large sample size (n), the distribution of sample statistics for a mean ($\bar{x}$) or a proportion ($\hat{p}$) is:

- Normally distributed
- Centered at the value of the population parameter



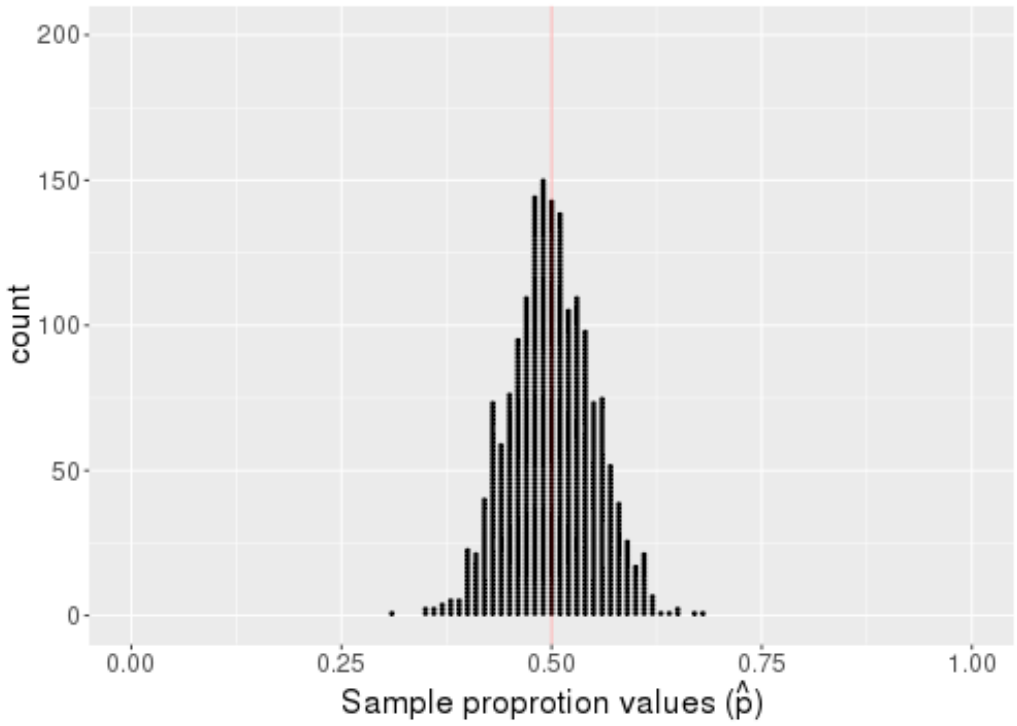
Upshot: We can create confidence intervals and run hypothesis tests using the normal distribution

- Rather than using computational methods like the bootstrap or a randomization methods to create a null distribution!

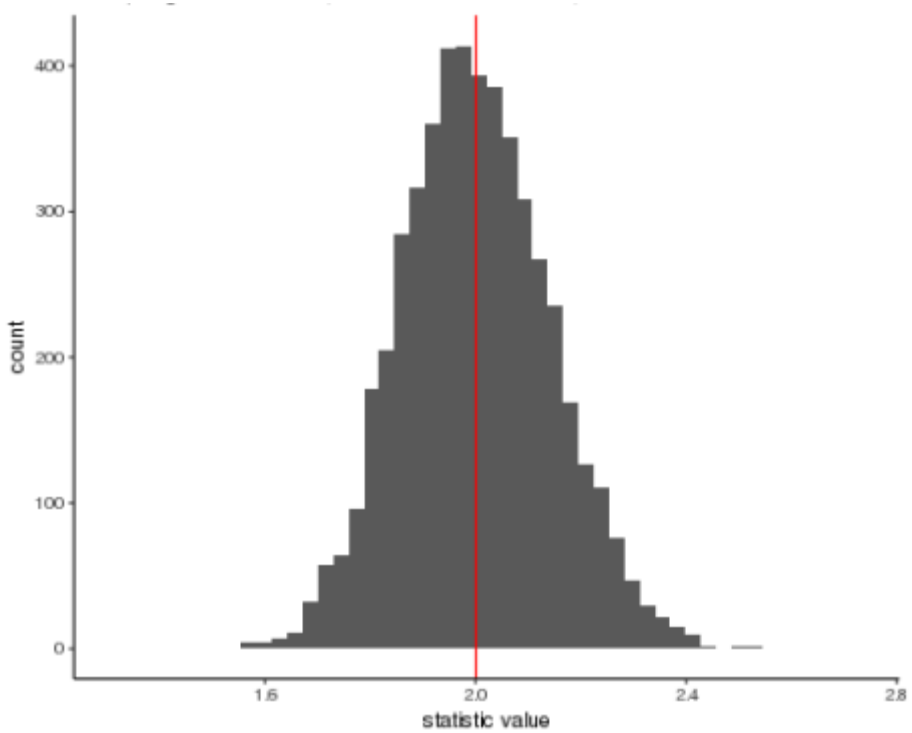


Central limit theorem

proportion ($\hat{p}$)



mean ($\bar{x}$)



[Proportion sampling distribution app]

[Sampling/Bootstrap distribution app]

The plan

For large n, the sampling distributions of $\bar{x}$ and $\hat{p}$ have normal distributions

We can convert any normal distribution $N(\mu, \sigma)$, into a standard normal distribution $N(0, 1)$

We can then use the standard normal distribution for inference

- I.e., To create confidence intervals and run hypothesis tests